SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305
Name:T Gnaneswari
Reg no:192372183

COURSE OUTCOME COVERED

CO 5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems. (L4, L5)

Assessment Tool: Viva Voce

Weightage: 10%

QUESTIONS: **Total Marks: 20**

1. Define SOAP.
2. Define WSDL.
3. What is an API?
4. Define XML Schema.
5. What is a Web Service?
6. Define REST.
7. What is a Protocol?
8. Define Endpoint in web services.
9. What is a Service in web applications?
10. What is the difference between SOAP and REST?
11. What is the role of WSDL in web services?
12. What is JSON and how is it used in APIs?
13. What is HTTP and why is it used in web services?
14. What is meant by request and response in web communication?
15. What is the importance of XML in web services?
16. What is the purpose of an API endpoint?
17. What is the difference between HTTP and HTTPS?
18. What is meant by RESTful services?
19. What is a namespace in XML?
20. What is the role of a client and server in web services?

Code of Conduct:

I T Gnaneswari(192372183)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

T Gnaneswari

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrate d
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively

1. Define SOAP.

SOAP (Simple Object Access Protocol) is a protocol used for exchanging structured information between applications over a network. It commonly uses XML for message formatting.

2. Define WSDL.

WSDL (Web Services Description Language) is an XML-based language used to describe a web service, including its operations, messages, data types, and endpoint.

3. What is an API?

API (Application Programming Interface) is a set of rules that allows different software applications to communicate and exchange data or services.

4. Define XML Schema.

XML Schema (XSD) defines the structure, elements, attributes, and data types allowed in an XML document. It is used to validate XML data.

5. What is a Web Service?

A Web Service is a software system that allows different applications to communicate and exchange data over a network, usually using standard web protocols.

6. Define REST.

REST (Representational State Transfer) is an architectural style for designing web services. It uses standard HTTP methods such as GET, POST, PUT, and DELETE to work with resources.

7. What is a Protocol?

A protocol is a set of rules and standards that define how data is transmitted and exchanged between devices or applications.

Example: HTTP, HTTPS, FTP, TCP/IP.

8. Define Endpoint in web services.

An endpoint is a specific URL/address through which a client can access a web service or API.

Example: https://example.com/api/users

9. What is a Service in web applications?

A service is a software component that provides a specific function or functionality to other applications or users over a network.

Example: A payment service processes online payments.

10. What is the difference between SOAP and REST?

SOAP	REST
SOAP is a protocol	REST is an architectural style
Mainly uses XML	Commonly uses JSON, XML, etc.
More complex	Simpler and lightweight
Uses WSDL for service description	Usually uses OpenAPI or documentation
Supports built-in standards such as WS-Security	Relies mainly on HTTP security mechanisms
Suitable for enterprise applications	Commonly used for web and mobile APIs

11. What is the role of WSDL in web services?

WSDL describes how a web service can be accessed and used. It specifies:

- Available operations
- Input and output messages
- Data types
- Communication protocol
- Service endpoint

Thus, WSDL helps a client understand how to communicate with a SOAP web service.

12. What is JSON and how is it used in APIs?

JSON (JavaScript Object Notation) is a lightweight data-interchange format.

It is commonly used in APIs to send and receive data between clients and servers.

Example:

```
{
  "name": "John",
  "age": 21
}
```

13. What is HTTP and why is it used in web services?

HTTP (HyperText Transfer Protocol) is a communication protocol used to transfer data over the web.

Web services use HTTP because it provides standard methods such as:

- GET – retrieve data

- POST – create/send data
- PUT – update data
- DELETE – remove data

14. What is meant by request and response in web communication?

A request is a message sent by the client to the server asking for a resource or operation.

A response is the message sent by the server back to the client containing the requested data or the result.

Example:

Client → GET request → Server

Server → JSON response → Client

15. What is the importance of XML in web services?

XML is important because it provides a standard, structured, and platform-independent way of representing data.

It is widely used in SOAP messages and WSDL documents.

16. What is the purpose of an API endpoint?

An API endpoint provides a specific URL through which a client can access a particular resource or operation.

Example:

GET /api/students → retrieves student information.

17. What is the difference between HTTP and HTTPS?

HTTP	HTTPS
HyperText Transfer Protocol	HyperText Transfer Protocol Secure
Data is generally transmitted without encryption	Data is encrypted
Less secure	More secure
Uses HTTP	Uses HTTP over TLS
Suitable for non-sensitive communication	Recommended for sensitive data

18. What is meant by RESTful services?

RESTful services are web services designed according to REST principles. They use HTTP methods to perform operations on resources.

For example:

- GET /students → Get students
- POST /students → Add a student
- PUT /students/1 → Update student 1
- DELETE /students/1 → Delete student 1

19. What is a namespace in XML?

An XML namespace is used to uniquely identify XML elements and avoid name conflicts when elements from different XML vocabularies are used together.

Example:

```
<book xmlns="http://example.com/books">
  <title>Web Services</title>
</book>
```

20. What is the role of a client and server in web services?

- Client: Sends requests to access a service or resource.
- Server: Receives requests, processes them, and sends responses.

Example:

Client → Request → Server → Processing → Response → Client

In short: The client requests a service, and the server provides the service.